06 The Coexistence Room

Competitive exclusion & niche partitioning

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 35 min	**+ 30 min**	The Resource Room (03) for R^*, and the Neighbours Room (05) for the outcomes.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Coexistence tab

Two consumers eat the same tracked food, so competition happens for a reason rather than through a coefficient. In one-resource mode there is a single supply; in two-resource mode each species can specialise. This guide uses the R^* table to predict the winner before the run.

LEARNING OBJECTIVES

After the core path you should be able to

- Use $R^* = m/(ea)$ to predict the winner of competition for a single resource.
- Show that starting abundance does not change that outcome.
- Explain how two resources make coexistence possible (niche partitioning).
- Explain why coexistence collapses when both species are limited by the same resource.

The extension adds

- Calculate a full R^* table and use it to design a reversal of the outcome.
- Read the *Who is limited by what* panel as a Tilman diagram.
- Predict how the ratio of resource supply decides which species persist.

SUGGESTED ROUTES

- **40-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Paired with the Neighbours Room:** the core paths of both, compared through question 3.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 One resource, two consumers

10 min

SET UP

One resource. The defaults: A has a = 0.020, m = 0.08; B has a = 0.030, m = 0.30; e = 0.40; both start at 10.

PREDICT FIRST

B eats faster. Calculate R^* for each species. Which will win?

RUN

Press **Run**.

LOOK AT

the *Break-even table (R)** and the dashed R^* lines on *Resources over time*.

EXPLAIN

Why does a species that eats faster still lose?

2 A head start

8 min

SET UP

As before, but B starts at **100** and A at 10.

PREDICT FIRST

Will a tenfold head start save B?

RUN

Run.

EXPLAIN

Why does starting abundance not matter here? Compare with founder control in the Neighbours Room.

3 Two resources

10 min

SET UP

Two resources, defaults: A attacks R1 at 0.020 and R2 at 0.005; B attacks R1 at 0.030 and R2 at 0.050.

PREDICT FIRST

Each species is better on a different resource relative to the other. Will both persist?

RUN

Run.

LOOK AT

both resource curves and the *Who is limited by what* panel.

EXPLAIN

Explain the coexistence in terms of which resource each species depletes and which limits it.

4 Both want the same thing

7 min

SET UP
Two resources; set B's attack on R2 to **0.005**, so that both species now do best on R1.

PREDICT FIRST
Will they still coexist?

RUN
Run.

EXPLAIN
Why does coexistence need each species to be limited by a different resource?

EXTENSION · MSC

5 Reversing the outcome

10 min

SET UP
One resource, defaults.

PREDICT FIRST
How low must B's mortality be for B to win? Calculate, then choose a value.

RUN
Set m_B and run.

EXPLAIN
Which three parameters could each, alone, reverse the outcome? Why is the starting abundance not one of them?

6 Reading the Tilman diagram

12 min

SET UP
Two resources, defaults.

PREDICT FIRST
In the *Who is limited by what* panel each species' break-even condition is a line in resource space. Predict where the two lines cross and where the resources will settle.

RUN
Run and compare.

EXPLAIN
State the two conditions for stable coexistence on two substitutable resources: the lines must cross, and each species must consume relatively more of the resource that limits it more.

7 Supply decides

8 min

SET UP

Two resources, defaults; then S_1 = **30**, S_2 = **4**; then S_1 = **4**, S_2 = **30**.

PREDICT FIRST

Which species will persist in each case?

RUN

Run all three.

EXPLAIN

Why can the same two species coexist in one environment and not in another? What does this predict about diversity along a gradient of nutrient ratios?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Why is R^* sometimes called "how hungry a species can stay alive"?
2. BSc Phytoplankton species in a lake compete for nitrogen and phosphorus. Using this room, explain how many species could coexist at equilibrium on two resources.
3. BSc Compare how this room and the Neighbours Room explain coexistence. Which one tells you *why*?
4. MSc Hutchinson's "paradox of the plankton" asks how so many species coexist on few resources. Give two resolutions, and say which this room excludes.
5. MSc The resources here are substitutable. How would essential resources (both needed) change the geometry?
6. MSc Tilman's experiments with diatoms tested R^* predictions. Summarise the design and result.

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** $R_A = 0.08/(0.4 \times 0.02) = 10$; $R_B = 0.3/(0.4 \times 0.03) = 25$. A wins by drawing the resource below what B needs.
- **Step 2.** A still wins: B drives the resource down quickly at first, but once it falls below 25 B can only decline.
- **Step 3.** Both persist (A near 54, B near 15): each is held back mainly by the resource it uses relatively more.
- **Step 4.** B is excluded and A takes over: with both limited by R1, the species with the lower R^* on it wins.
- **Step 5.** B wins if $m_B < 0.4 \times 0.03 \times 10 = 0.12$. Mortality, attack rate or efficiency can reverse it; starting abundance cannot.
- **Step 6.** The lines cross, and the resources settle near the crossing; the consumption ratios meet the coexistence condition.
- **Step 7.** Defaults: both persist. $S_1 = 30$, $S_2 = 4$: only A. $S_1 = 4$, $S_2 = 30$: only B. The supply ratio picks the winner, so a gradient of ratios can hold more species than any one site.